**D 31616** **(Pages : 5)** **Name.....**

**Reg. No.....**

**THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION**
**NOVEMBER 2022**

Mathematics

MAT 3C 03—MATHEMATICS

(2017—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

**Part A (Objective Type)**

*Answer all the* **twelve** *questions.*
*Each question carries* 1 *mark.*

1. Define partial differential equation.

2. The solution of the differential equation $y' = -xy$ is ————————.

3. The degree of the differential equation $\frac{dy}{dx} = \cot x$ is ————————.

4. The rank of the matrix $A = \begin{bmatrix} 0 & 2 \\ 3 & 5 \end{bmatrix}$ is ————————.

5. State True *or* False : The following two matrices are equivalent :

$$\begin{bmatrix} 7 & 5 & -5 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 3 & -3 & 3 & -7 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 0 & -2 & -4 \\ 3 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 3 & -3 & 3 & -7 \end{bmatrix}.$$

6. The characteristic matrix of the matrix $\begin{bmatrix} 5 & 3 \\ 4 & 2 \end{bmatrix}$ is ————————.

7. Find the resultant of the vectors $\mathbf{p}$, $\mathbf{q} = 3\mathbf{p}$, and $\mathbf{u}$, where $\mathbf{p} = [3, 5, 7]$, $\mathbf{q} = 3\mathbf{p}$, and $\mathbf{u} = -6\mathbf{p}$.

8. If $\mathbf{a} = [1, 2, 0]$, $\mathbf{b} = [-3, 2, 0]$, and $\mathbf{c} = [2, 3, 4]$, then $(\mathbf{a} + \mathbf{b}) \times \mathbf{c}$.

**Turn over**

9. The vector function $\mathbf{r}(t) = 5\cos t\,\mathbf{i} + 7\sin t\,\mathbf{j}\,(0 \le t \le 2\pi)$ represents ————————.

10. A particle moves along the curve :

$x = 3t^2,\ y = t^2 - 2t,\ z = t^3$.

Find the velocity at $t = 1$.

11. If $r = \dfrac{(-y\mathbf{i} + x\mathbf{j})}{x^2 + y^2}$ then div $\mathbf{v}$ = ————————.

12. Prove that curl $\mathbf{r} = 0$.

(12 × 1 = 12 marks)

**Part B (Short Answer Type)**

*Answer any* **nine** *questions.*
*Each question carries* 2 *marks.*

13. Verify that $y = e^{5x}$ is a solution of the differential equation $\dfrac{dy}{dx} = 5y$.

14. Solve the initial value problem $(x + 1)\,y' - 2y = 0,\ y\,(0) = 1$.

15. Show that the equation :

$xdy - ydx = 0$

is exact and solve it.

16. Reduce the matrix $\begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}$ to its normal form.

17. Solve completely the system of equations :

$$\begin{aligned} x + 3y - 2z &= 0 \\ 2x - y + 4z &= 0 \\ x - 11y + 14z &= 0. \end{aligned}$$

18. Find the eigen values of :

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & -1 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

19. Prove that $(\mathbf{u} \cdot \mathbf{v})' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$.

20. Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at any point P: (2, 1, 3) in the direction of the vector $\mathbf{a} = \mathbf{i} - 2\mathbf{k}$.

21. Determine whether $\mathbf{F}(x, y) = 4y\mathbf{i} + 4x\mathbf{j}$ is a conservative vector field.

22. Find a tangent vector $r'(t)$ at the point $\left(\frac{1}{2}, \sqrt{3}, 0\right)$ to the curve $\mathbf{r}(t) = \cos t\, \mathbf{i} + 2 \sin t\, \mathbf{j}$.

23. Find unit normal to the surface $x^2y + 2xz = 4$ at the point (2, – 2, 3).

24. Verify that $w = x^2 - y^2$ satisfies Laplace's equation $\nabla^2 w = 0$.

(9 × 2 = 18 marks)

## Part C (Short Essays)

*Answer any* **six** *questions.*
*Each question carries 5 marks.*

25. Solve the initial value problem $y' - y = e^x$, $y(1) = 0$.

26. Determine the rank of the following matrix, by reducing to echelon form :

$$\begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix}.$$

27. Test for consistency and solve the system :

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 2 \\ 3x_1 + x_2 - 2x_3 &= 1 \\ 4x_1 - 3x_2 - x_3 &= 3 \\ 2x_1 + 4x_2 + 2x_3 &= 4. \end{aligned}$$

28. Find the eigen values and the eigen vector corresponding to the largest eigen value of the matrix

$$\begin{bmatrix} 3 & 0 & 0 \\ 5 & 4 & 0 \\ 3 & 6 & 1 \end{bmatrix}.$$

29. Verify the Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$.

30. Find the work done in moving a particle once round a circle C in the $xy$ plane : the circle has centre at the origin at radius 3 and the force field is given by :

$$\mathbf{F} = (2x - y + z)\mathbf{i} + \left(x + y - z^2\right)\mathbf{j} + (3x - 2y + 4z)\mathbf{k}.$$

31. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$ by the divergence theorem where $\mathbf{F} = \left[x^2, 0, z^2\right]$ and S is the surface of the box given by the inequalities $|x| \le 1, |y| \le 3, |z| \le 2$.

32. Evaluate the flux integral $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$ where

$\mathbf{F} = \left[3x^2, y^2, 0\right]$, and $S : \mathbf{r} = [u, v, 2u + 3v], 0 \le u \le 2, -1 \le v \le 1$.

33. Use Green's theorem to evaluate $\oint_C y dx + x dy$, where C is the unit circle oriented counterclockwise.

(6 × 5 = 30 marks)

## Part D

*Answer any* **two** *questions.*
*Each question carries 10 marks.*

34. Find the orthogonal trajectory of the family of circles $(x-c)^2 + y^2 = c^2$.

35. Investigate for what values of $a$, $b$ the system of equations :

$$\begin{aligned} x + y + 2z &= 2 \\ 2x - y + 3z &= 10 \\ 5x - y + az &= b \end{aligned}$$

have an infinite number of solutions.

36. Calculate the line integral $\oint_C \mathbf{F} \cdot \mathbf{r}'(s)\, ds$ using Stoke's theorem, where F = [4$z$, – 2$x$, 2$x$], and C is the ellipse $x^2 + y^2 = 1, z = y + 1$.

(2 × 10 = 20 marks)